

Deterministic Optimization

Robust Optimization

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More Examples of Robust Linear
Optimization

Robust Optimization

Learning Objectives for this lesson

- Explore a new way to model uncertainty

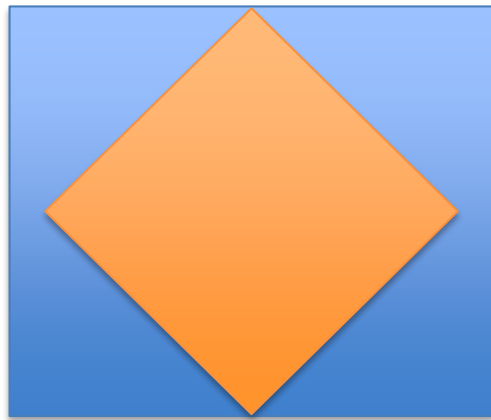
Budgeted Uncertainty Set

- Robust linear constraint
 - $a^\top x \geq b \quad \forall a \in U$
 - Coefficients of $a = (a_1, \dots, a_n)$ are uncertain
 - For example, a_i is the return of stock i in the next month
 - $x = (x_1, \dots, x_n)$ is the investment decision for stocks $i = 1, \dots, n$
 - The return a_i is estimated to be in the interval $a_i \in [\bar{a}_i - \hat{a}_i, \bar{a}_i + \hat{a}_i]$
 - It is reasonable to assume that not all the returns will take their lowest estimated value $\bar{a}_i - \hat{a}_i$ simultaneously. How to model this?
 - Introduce a **budget of uncertainty** Γ :

$$U = \left\{ a = (a_1, \dots, a_n) : \sum_i \frac{|a_i - \bar{a}_i|}{\hat{a}_i} \leq \Gamma, a_i \in [\bar{a}_i - \hat{a}_i, \bar{a}_i + \hat{a}_i], \forall i \right\}$$

Intuitive Understanding

$$U = \left\{ a = (a_1, \dots, a_n): \sum_i \frac{|a_i - \bar{a}_i|}{\hat{a}_i} \leq \Gamma, a_i \in [\bar{a}_i - \hat{a}_i, \bar{a}_i + \hat{a}_i], \forall i \right\}$$



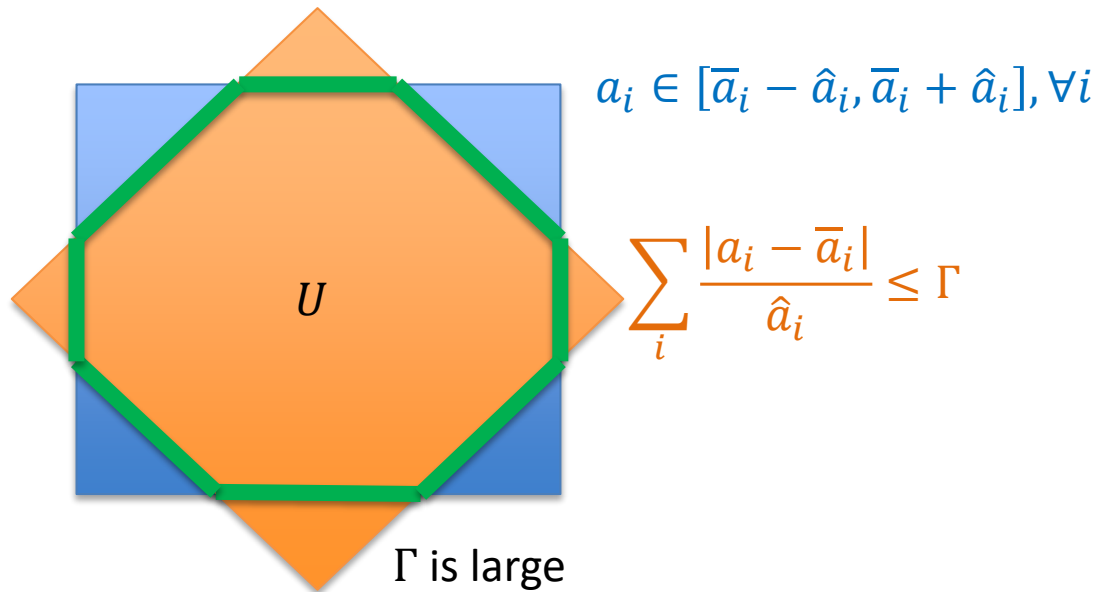
$$a_i \in [\bar{a}_i - \hat{a}_i, \bar{a}_i + \hat{a}_i], \forall i$$

$$\sum_i \frac{|a_i - \bar{a}_i|}{\hat{a}_i} \leq \Gamma$$

Γ is small

Intuitive Understanding

$$U = \left\{ a = (a_1, \dots, a_n): \sum_i \frac{|a_i - \bar{a}_i|}{\hat{a}_i} \leq \Gamma, a_i \in [\bar{a}_i - \hat{a}_i, \bar{a}_i + \hat{a}_i], \forall i \right\}$$



Reformulation of Robust Constraint

- Equivalent reformulation:
 - $a^\top x \geq b \quad \forall a \in U$ is equivalent to
 - $\min_{a \in U} a^\top x \geq b$

$$\begin{aligned} \min_a a^\top x &\geq b \\ \text{s.t. } \sum_i \frac{|a_i - \bar{a}_i|}{\hat{a}_i} &\leq \Gamma \\ -a_i &\leq -\bar{a}_i + \hat{a}_i \quad \forall i \\ a_i &\leq \bar{a}_i + \hat{a}_i \quad \forall i \end{aligned}$$



$$\begin{aligned} \min_{a, w} a^\top x &\geq b \\ \text{s.t. } \sum_i \frac{w_i}{\hat{a}_i} &\leq \Gamma \\ -a_i &\leq -\bar{a}_i + \hat{a}_i \quad \forall i \\ a_i &\leq \bar{a}_i + \hat{a}_i \quad \forall i \\ -w_i + a_i &\leq \bar{a}_i \quad \forall i \\ -w_i - a_i &\leq -\bar{a}_i \quad \forall i \end{aligned}$$

Reformulation (cont.)

- Use LP duality:

$$\begin{aligned} \min_{a,w} \quad & a^\top x && \geq b \\ \text{s.t.} \quad & \sum_i \frac{w_i}{\hat{a}_i} \leq \Gamma \\ & -a_i \leq -\bar{a}_i + \hat{a}_i \quad \forall i \\ & a_i \leq \bar{a}_i + \hat{a}_i \quad \forall i \\ & -w_i + a_i \leq \bar{a}_i \quad \forall i \\ & -w_i - a_i \leq -\bar{a}_i \quad \forall i \end{aligned}$$

$$\max \Gamma s + \sum_i [(-\bar{a}_i + \hat{a}_i)y_i^- + (\bar{a}_i$$

Reformulation (cont.)

- Use LP duality:

$$\begin{aligned} \min_{a,w} \quad & a^\top x && \geq b \\ \text{s.t.} \quad & \sum_i \frac{w_i}{\hat{a}_i} \leq \Gamma \\ & -a_i \leq -\bar{a}_i + \hat{a}_i \quad \forall i \\ & a_i \leq \bar{a}_i + \hat{a}_i \quad \forall i \\ & -w_i + a_i \leq \bar{a}_i \quad \forall i \\ & -w_i - a_i \leq -\bar{a}_i \quad \forall i \end{aligned}$$

$$\begin{aligned} \Gamma s + \sum_i [(-\bar{a}_i + \hat{a}_i)y_i^- + (\bar{a}_i + \hat{a}_i)y_i^+ \\ + \bar{a}_i(z_i^+ - z_i^-)] &\geq b \\ \frac{1}{\hat{a}_i} s - z_i^+ - z_i^- &= 0 \quad \forall i \\ y_i^+ - y_i^- + z_i^+ - z_i^- &= x_i \quad \forall i \\ s, y_i^-, y_i^+, z_i^-, z_i^+ &\leq 0 \quad \forall i \end{aligned}$$

Summary

- We learned:
 - The concept and intuition of budgeted uncertainty sets
 - Reformulation of robust constraint under budgeted uncertainty sets